SIDEBAR **Entropy versus standard deviation**

When we used the term "uncertainty" in previous chapters, we referred to the standard deviations of the predictive normal distributions produced by a Gaussian process. The standard deviation of a distribution, as the name suggests, measures how much the values within the distribution deviate from the mean and is therefore a valid measure of uncertainty.

Entropy, on the other hand, is information-theoretically motivated and is also a valid measure of uncertainty. In fact, it is a more elegant and general approach to quantify uncertainty, and could more accurately model uncertainty in the edge cases in many situations.

IMPORTANT For a given probability distribution, the entropy is defined to be $\sum p \log p$, where we sum over the different possible events indexed by i.

We see that what we used for the Bernoulli distribution above is a special case of this formula. We will also use this formula later in the chapter on uniform distributions.

6.1.2 Looking for a remote control using entropy

As entropy measures how much uncertainty there is in our knowledge about a quantity or event of interest, it could inform our decisions to help us most efficiently reduce our uncertainty about the quantity or event. We will discuss a toy example of this in this subsection, where we decide where to best look for a lost remove control. Albeit simple, the example presents the information-theoretic reasoning that we will use in subsequent discussions where entropy is used for more complex decision-making problems.

Imagine that while trying to turn on the TV in your living room on day, you realize that you can't find the remote control on the table, which is where it usually is. You therefore decide to conduct a search for this remote. First, you reason that it should be in the living room somewhere, but you don't have any idea about where the remote is within the living room, so all locations are equally likely. In the language of probabilistic inference that we've been using, you can say that the *distribution of the location of the remote* is uniform over the living room.

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